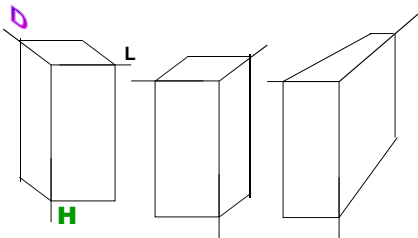
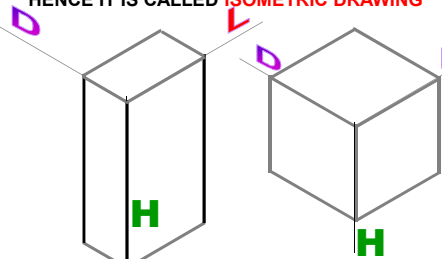


ISOMETRIC DRAWING IT IS A TYPE OF PICTORIAL PROJECTION IN WHICH ALL THREE DIMENSIONS OF AN OBJECT ARE SHOWN IN ONE VIEW AND IF REQUIRED, THEIR ACTUAL SIZES CAN BE MEASURED DIRECTLY FROM IT.	TYPICAL CONDITION. IN THIS 3-D DRAWING, AN OBJECT IS SO PLACED THAT, ITS THREE MUTUALLY PERPENDICULAR EDGES ARE EQUALLY INCLINED WITH THE PLANE OF PROJECTION. SO ALL THREE DIMENSIONAL AXES APPEAR AT EQUAL INCLINATIONS WITH EACH OTHER. (120°)
3-D DRAWINGS CAN BE DRAWN IN NUMEROUS WAYS AS SHOWN BELOW. ALL THESE DRAWINGS MAY BE CALLED 3-DIMENSIONAL DRAWINGS, OR PHOTOGRAPHIC OR PICTORIAL DRAWINGS. HERE NO SPECIFIC RELATION AMONG H, L & D AXES IS MAINTAINED. 	NOW OBSERVE BELOW GIVEN DRAWINGS. ONE CAN NOTE SPECIFIC INCLINATION AMONG H, L & D AXES. ISO MEANS SAME, SIMILAR OR EQUAL. HERE ONE CAN FIND EQUAL INCLINATION AMONG H, L & D AXES. EACH IS 120° INCLINED WITH OTHER TWO. HENCE IT IS CALLED ISOMETRIC DRAWING 
PURPOSE OF ISOMETRIC DRAWING IS TO UNDERSTAND OVERALL SHAPE, SIZE & APPEARANCE OF AN OBJECT PRIOR TO IT'S PRODUCTION.	

SOME IMPORTANT TERMS:

ISOMETRIC AXES, LINES AND PLANES:

The three lines AL, AD and AH, meeting at point A and making 120° angles with each other are termed **Isometric Axes**.

The lines parallel to these axes are called **Isometric Lines**.

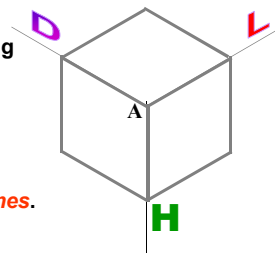
The planes representing the faces of the cube as well as other planes parallel to these planes are called **Isometric Planes**.

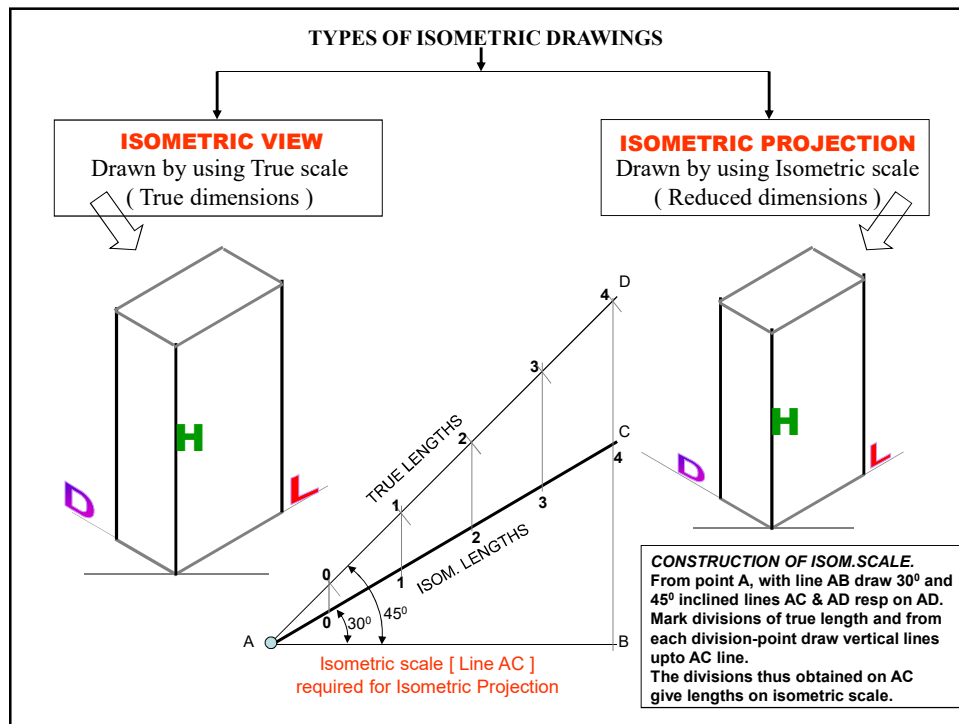
ISOMETRIC SCALE:

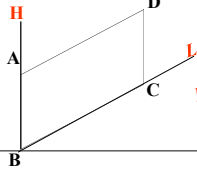
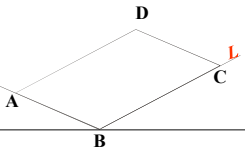
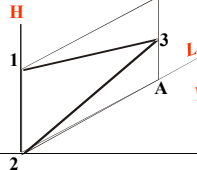
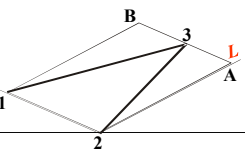
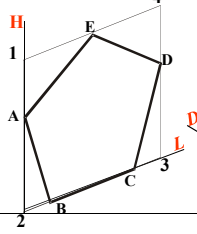
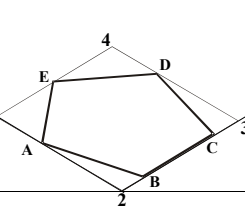
When one holds the object in such a way that all three dimensions are visible then in the process all dimensions become proportionally inclined to observer's eye sight and hence appear apparent in lengths.

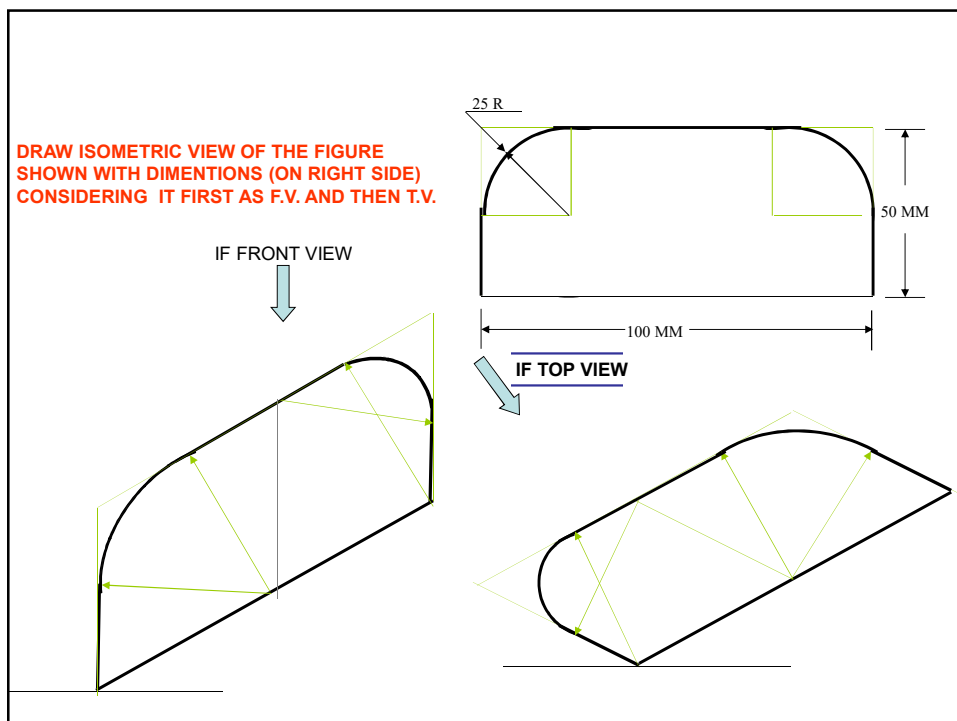
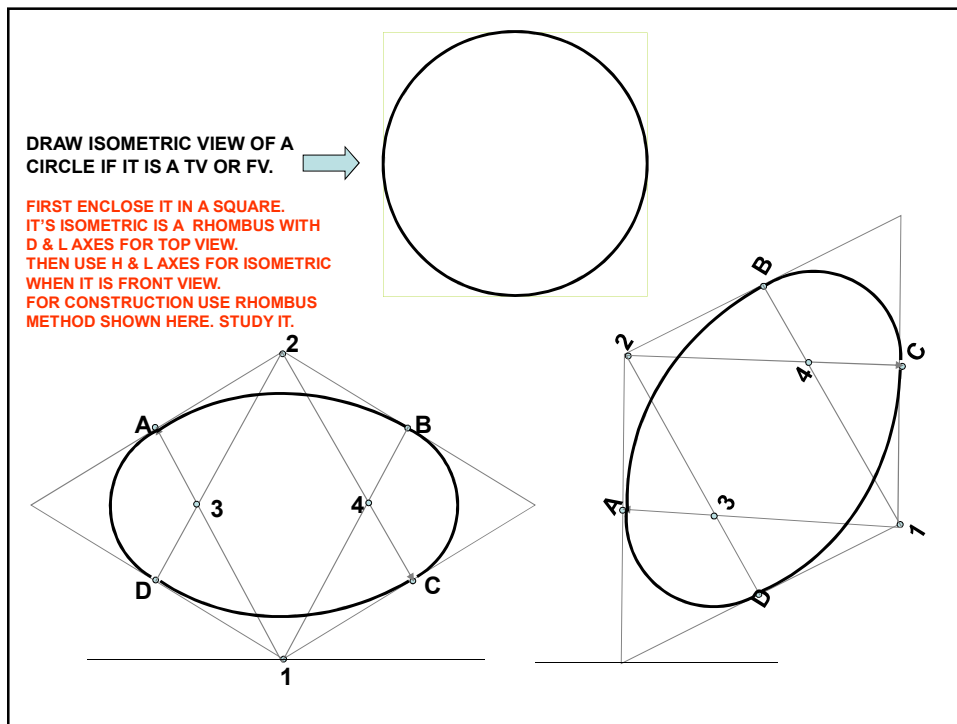
This reduction is 0.815 or 9 / 11 (approx.) It forms a reducing scale which is used to draw isometric drawings and is called **Isometric scale**.

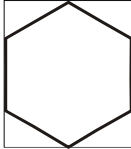
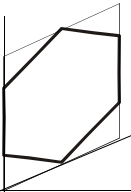
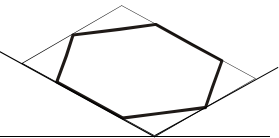
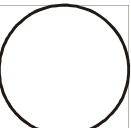
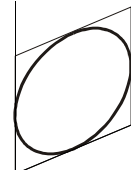
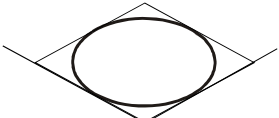
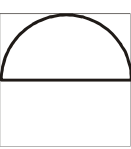
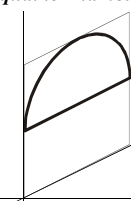
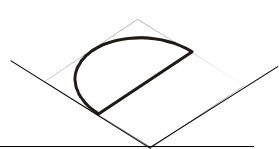
In practice, while drawing isometric projection, it is necessary to convert true lengths into isometric lengths for measuring and marking the sizes. This is conveniently done by constructing an isometric scale as described on next page.



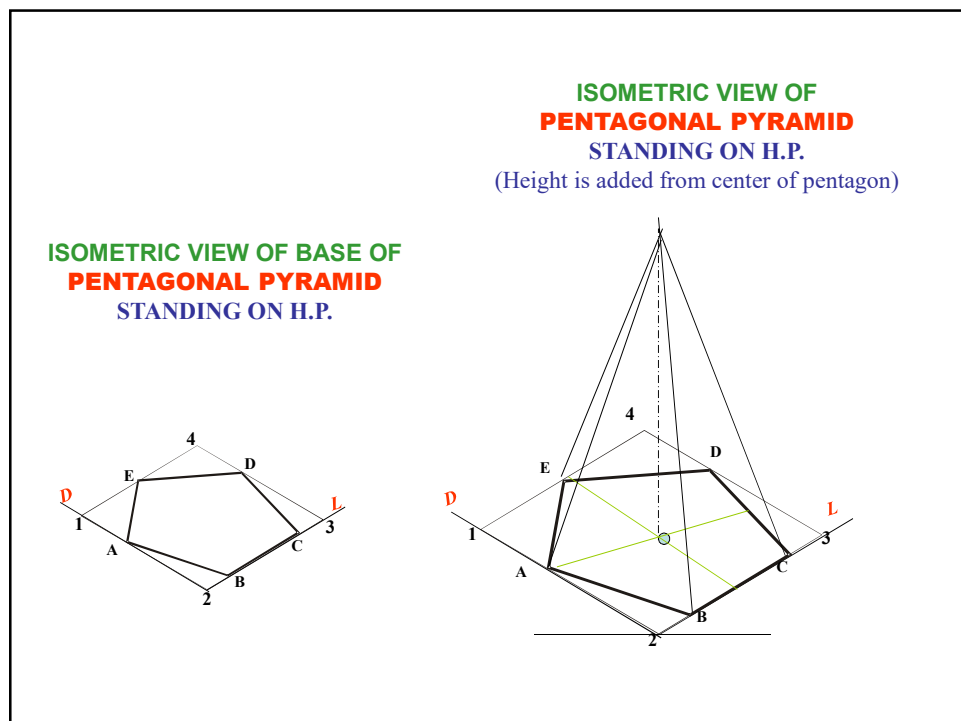


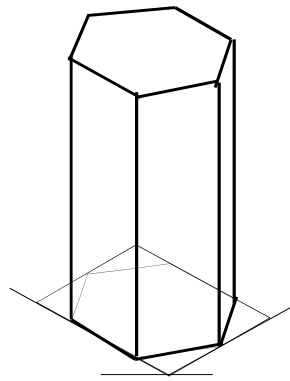
ISOMETRIC OF PLANE FIGURES	SHAPE	Isometric view if the Shape is F.V. or T.V.	
AS THESE ALL ARE 2-D FIGURES WE REQUIRE ONLY TWO ISOMETRIC AXES. IF THE FIGURE IS FRONT VIEW, H & L AXES ARE REQUIRED. IF THE FIGURE IS TOP VIEW, D & L AXES ARE REQUIRED. Shapes containing Inclined lines should be enclosed in a rectangle as shown. Then first draw isom. of that rectangle and then inscribe that shape as it is.	RECTANGLE		
	TRIANGLE		
	PENTAGON		



ISOMETRIC OF PLANE FIGURES	SHAPE	IF F.V.	IF T.V.
AS THESE ALL ARE 2-D FIGURES WE REQUIRE ONLY TWO ISOMETRIC AXES. IF THE FIGURE IS FRONT VIEW, H & L AXES ARE REQUIRED. IF THE FIGURE IS TOP VIEW, D & L AXES ARE REQUIRED. For Isometric of Circle/Semicircle use Rhombus method. Construct it of sides equal to diameter of circle always. (Ref. Previous two pages.)	HEXAGON 		
	CIRCLE 		
	SEMI CIRCLE 		

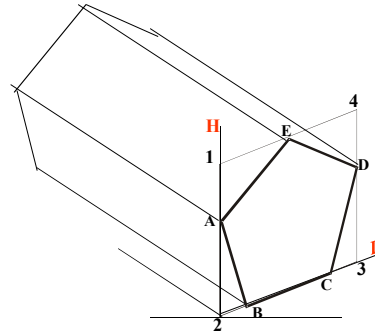
For Isometric of Circle/Semicircle use **Rhombus method**. Construct Rhombus of sides equal to Diameter of circle always.



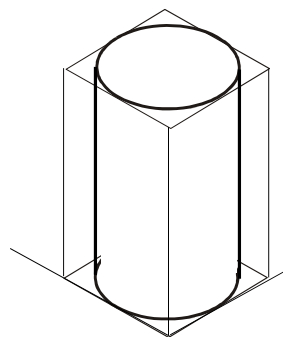


**ISOMETRIC VIEW OF
HEXAGONAL PRISM
STANDING ON H.P.**

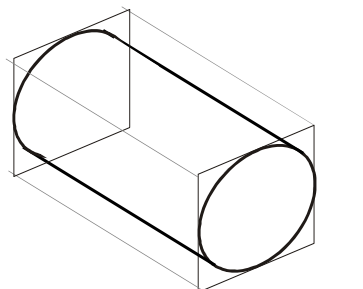
**ISOMETRIC VIEW OF
PENTAGONAL PRISM
LYING ON H.P.**



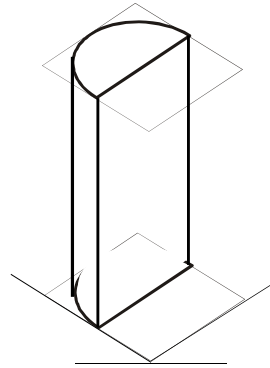
CYLINDER STANDING ON H.P.



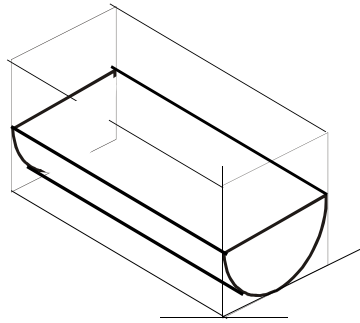
CYLINDER LYING ON H.P.



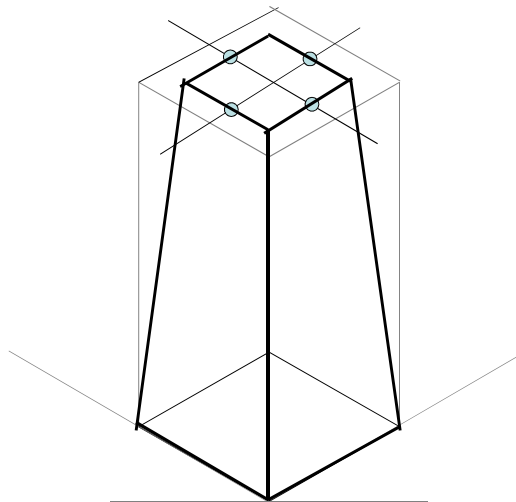
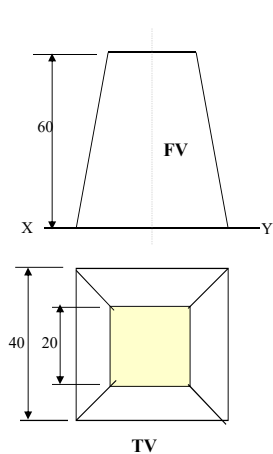
**HALF CYLINDER
STANDING ON H.P.
(ON IT'S SEMICIRCULAR BASE)**



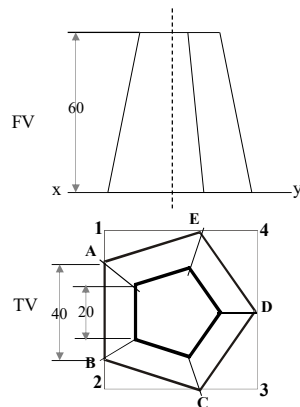
**HALF CYLINDER
LYING ON H.P.
(with flat face // to H.P.)**



**ISOMETRIC VIEW OF
A FRUSTUM OF SQUARE PYRAMID
STANDING ON H.P. ON IT'S LARGER BASE.**



**PROJECTIONS OF FRUSTUM OF
PENTAGONAL PYRAMID ARE GIVEN.
DRAW IT'S ISOMETRIC VIEW.**



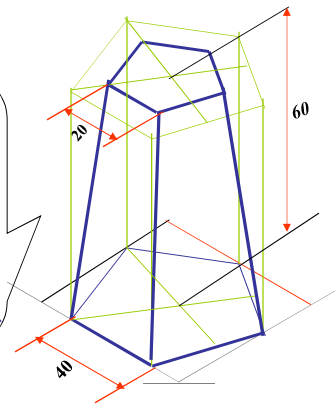
SOLUTION STEPS:

FIRST DRAW ISOMETRIC
OF IT'S BASE.

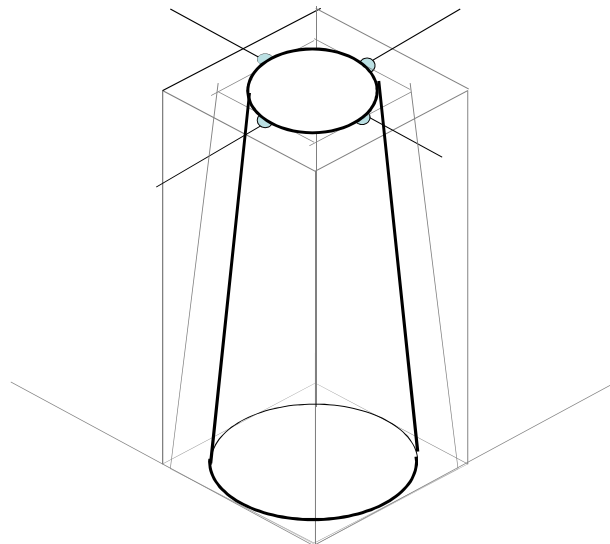
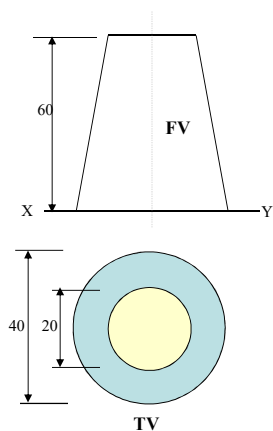
THEN DRAWSAME SHAPE
AS TOP, 60 MM ABOVE THE
BASE PENTAGON CENTER.

THEN REDUCE THE TOP TO
20 MM SIDES AND JOIN WITH
THE PROPER BASE CORNERS.

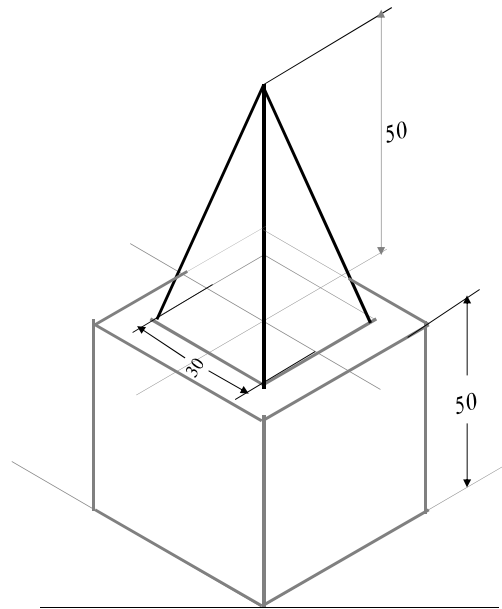
**ISOMETRIC VIEW
OF
FRUSTUM OF PENTAGONAL PYRAMID**



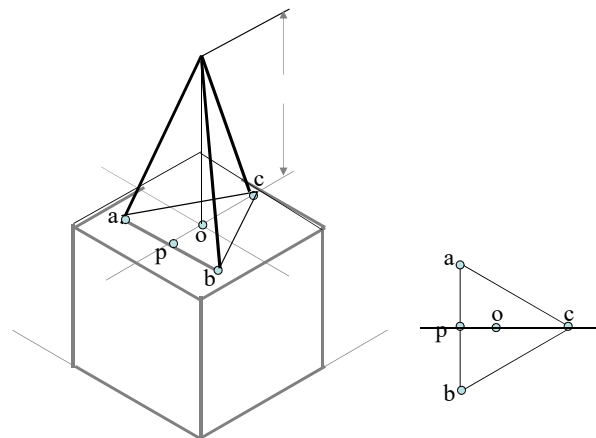
**ISOMETRIC VIEW OF
A FRUSTUM OF CONE
STANDING ON H.P. ON IT'S LARGER BASE.**



PROBLEM: A SQUARE PYRAMID OF 30 MM BASE SIDES AND 50 MM LONG AXIS, IS CENTRALLY PLACED ON THE TOP OF A CUBE OF 50 MM LONG EDGES. DRAW ISOMETRIC VIEW OF THE PAIR.



PROBLEM: A TRIANGULAR PYRAMID OF 30 MM BASE SIDES AND 50 MM LONG AXIS, IS CENTRALLY PLACED ON THE TOP OF A CUBE OF 50 MM LONG EDGES. DRAW ISOMETRIC VIEW OF THE PAIR.



SOLUTION HINTS.

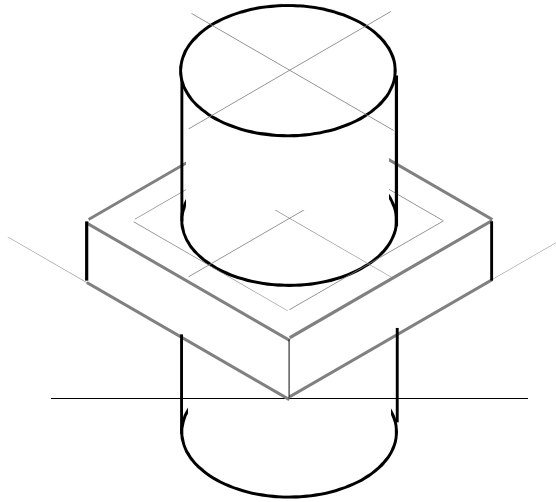
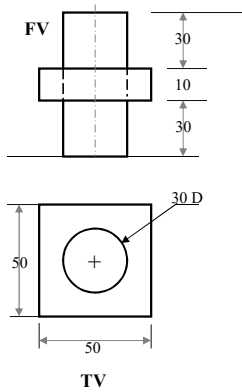
TO DRAW ISOMETRIC OF A CUBE IS SIMPLE. DRAW IT AS USUAL.

*BUT FOR PYRAMID AS IT'S BASE IS AN EQUILATERAL TRIANGLE,
IT CAN NOT BE DRAWN DIRECTLY. SUPPORT OF IT'S TV IS REQUIRED.*

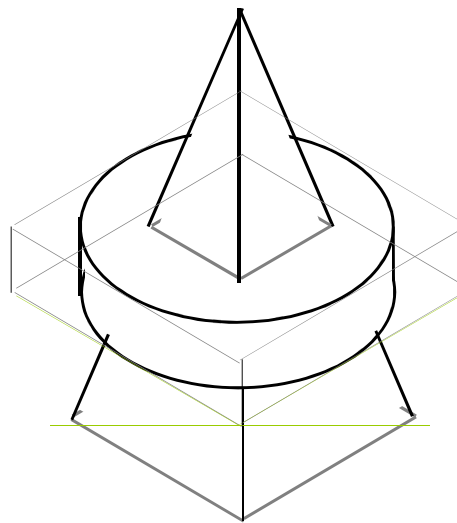
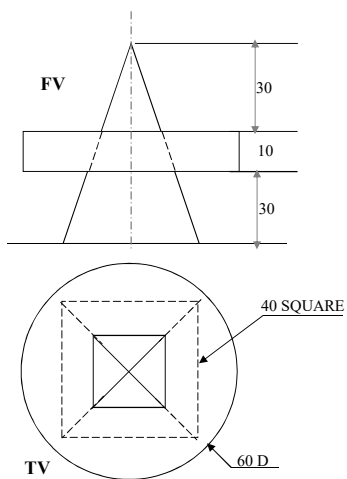
SO DRAW TRIANGLE AS A TV, SEPARATELY AND NAME VARIOUS POINTS AS SHOWN.
AFTER THIS PLACE IT ON THE TOP OF CUBE AS SHOWN.
THEN ADD HEIGHT FROM IT'S CENTER AND COMPLETE IT'S ISOMETRIC AS SHOWN.

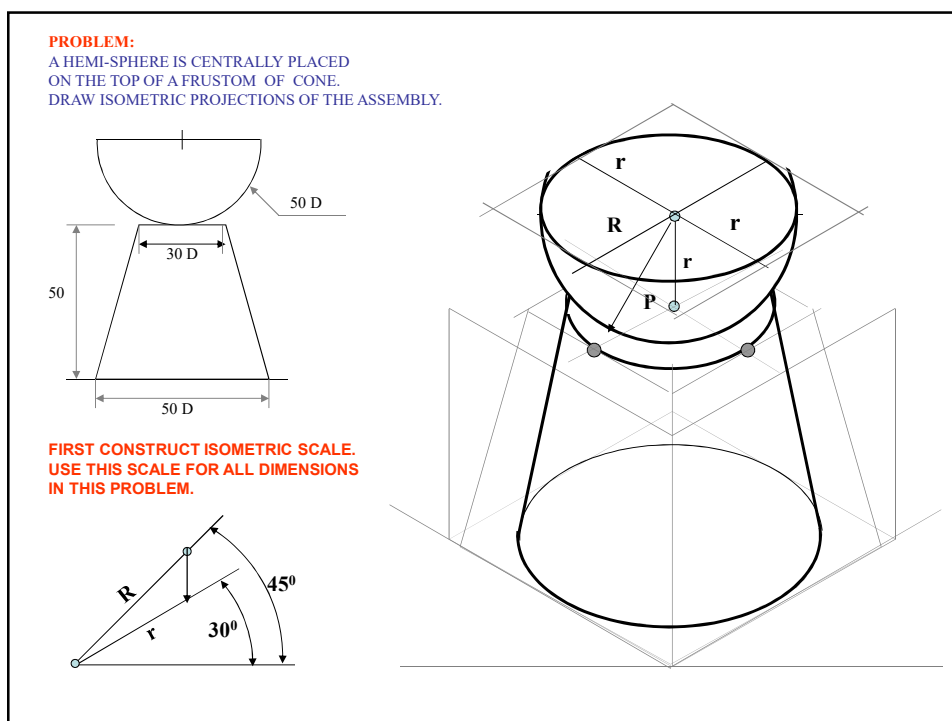
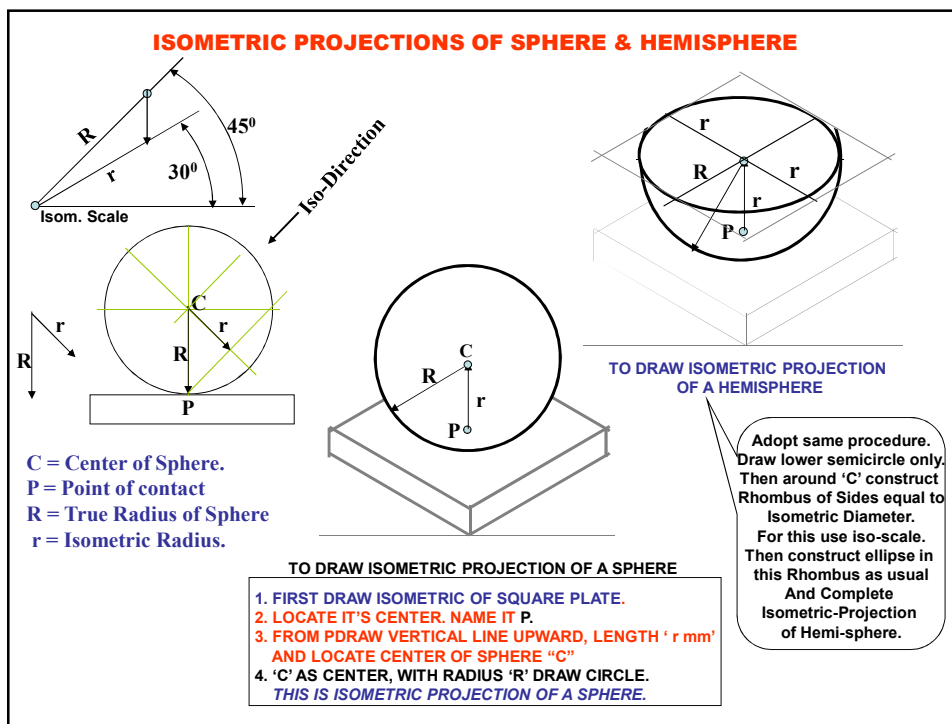
PROBLEM:

A SQUARE PLATE IS PIERCED THROUGH CENTRALLY BY A CYLINDER WHICH COMES OUT EQUALLY FROM BOTH FACES OF PLATE. IT'S FV & TV ARE SHOWN. DRAW ISOMETRIC VIEW.

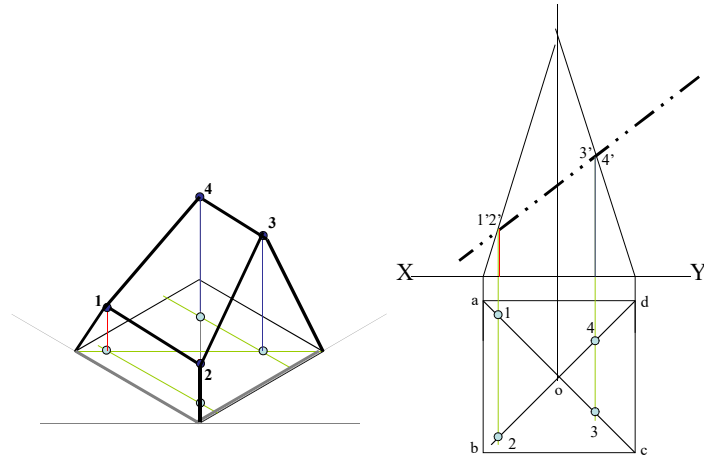
**PROBLEM:**

A CIRCULAR PLATE IS PIERCED THROUGH CENTRALLY BY A SQUARE PYRAMID WHICH COMES OUT EQUALLY FROM BOTH FACES OF PLATE. IT'S FV & TV ARE SHOWN. DRAW ISOMETRIC VIEW.

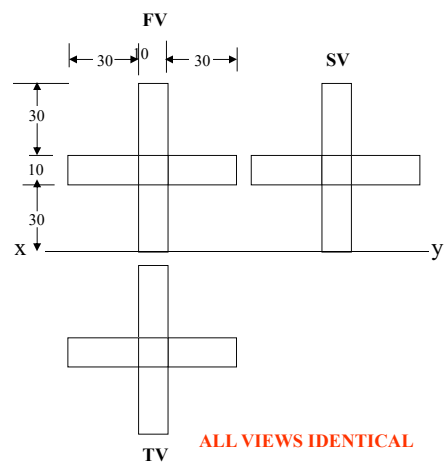
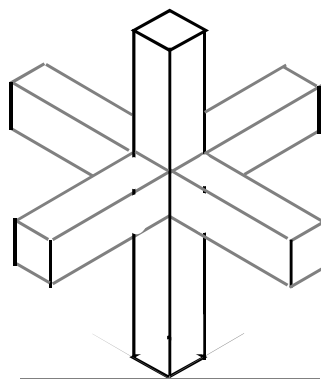




A SQUARE PYRAMID OF 40 MM BASE SIDES AND 60 MM AXIS IS CUT BY AN INCLINED SECTION PLANE THROUGH THE MID POINT OF AXIS AS SHOWN. DRAW ISOMETRIC VIEW OF SECTION OF PYRAMID.

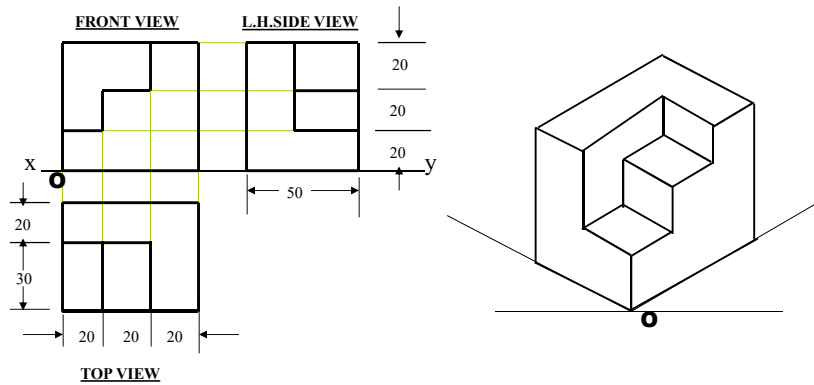


F.V. & T.V. and S.V. of an object are given. Draw its isometric view.

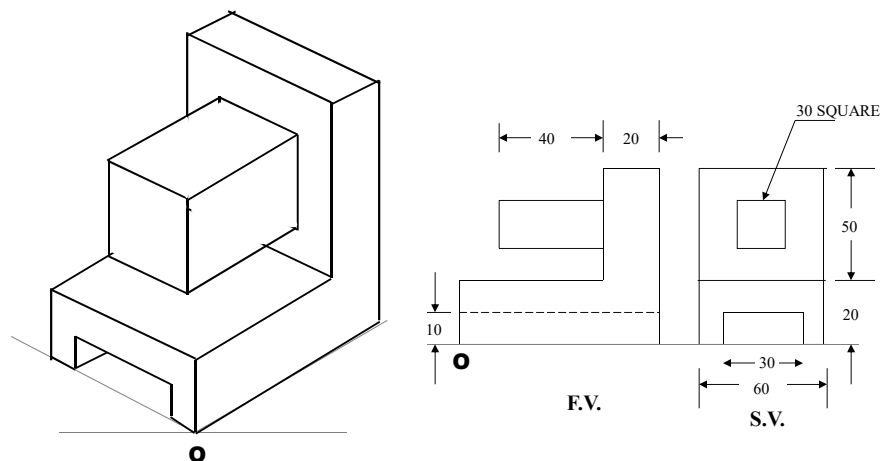


F.V. & T.V. and S.V. of an object are given. Draw its isometric view.

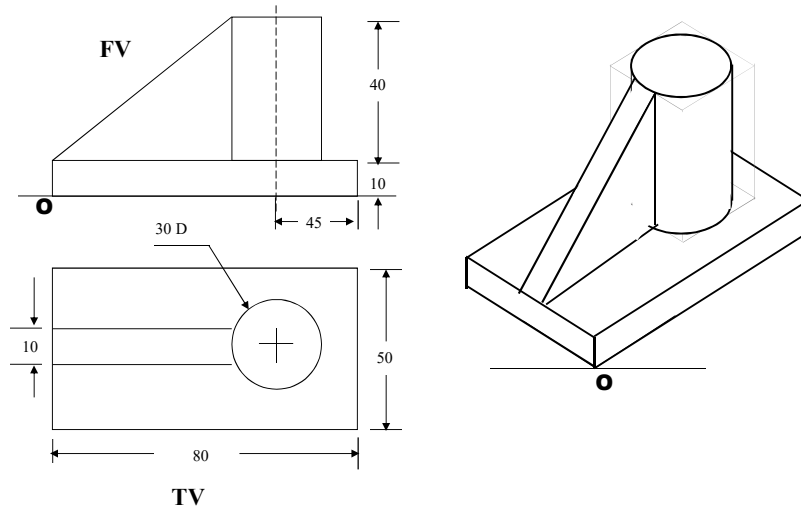
ORTHOGRAPHIC PROJECTIONS



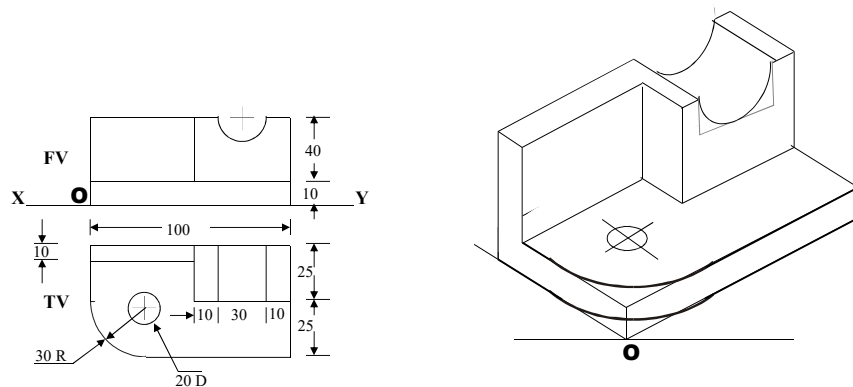
F.V. and S.V. of an object are given. Draw its isometric view.



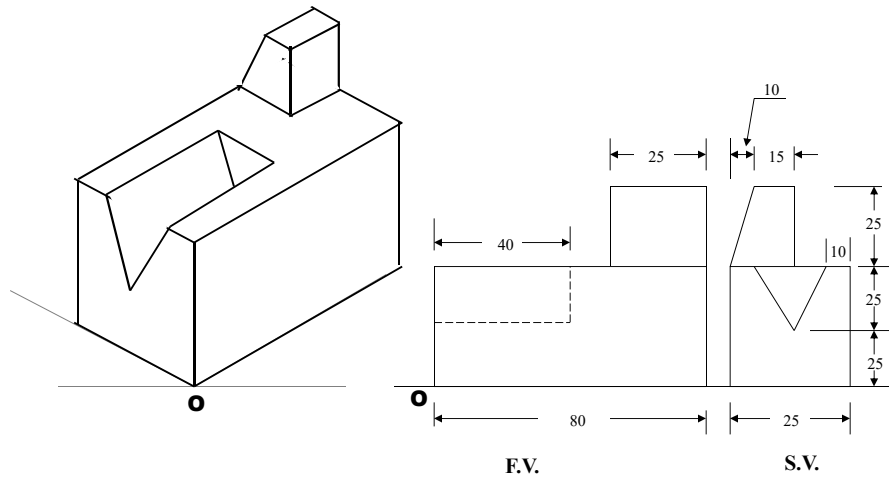
F.V. & T.V. of an object are given. Draw it's isometric view



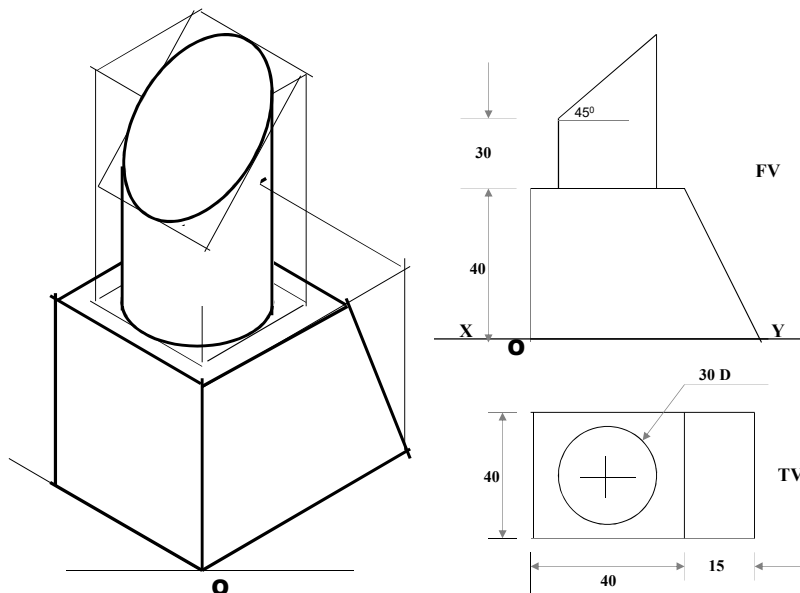
F.V. & T.V. of an object are given. Draw it's isometric view.



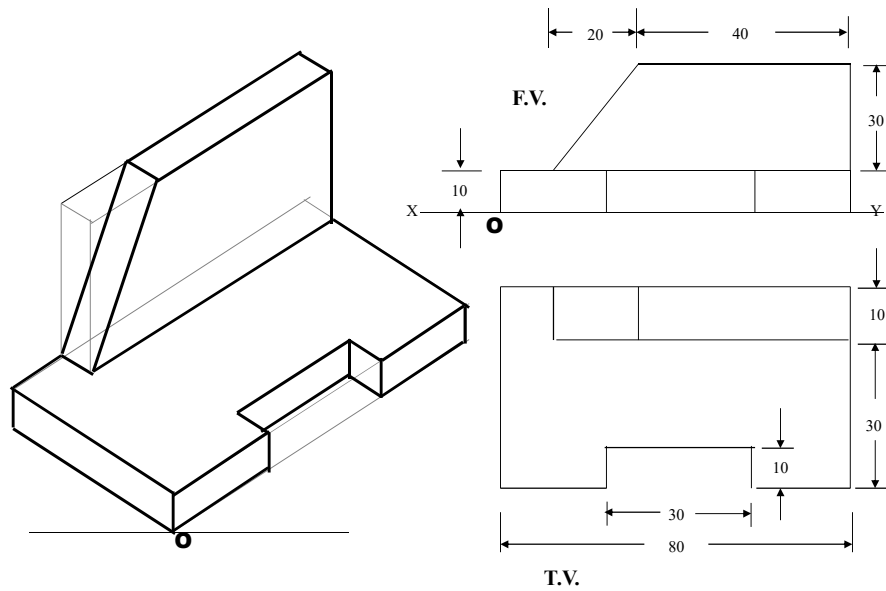
F.V. and S.V. of an object are given. Draw its isometric view.



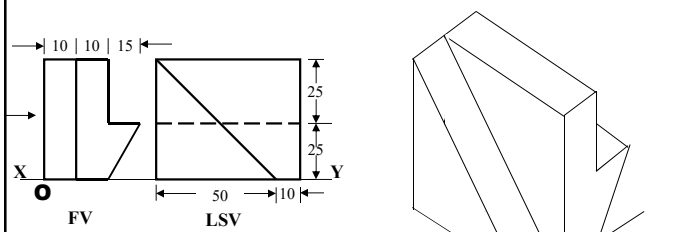
F.V. & T.V. of an object are given. Draw its isometric view.



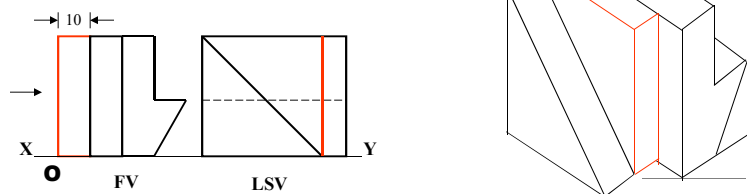
F.V. & T.V. of an object are given. Draw its isometric view.



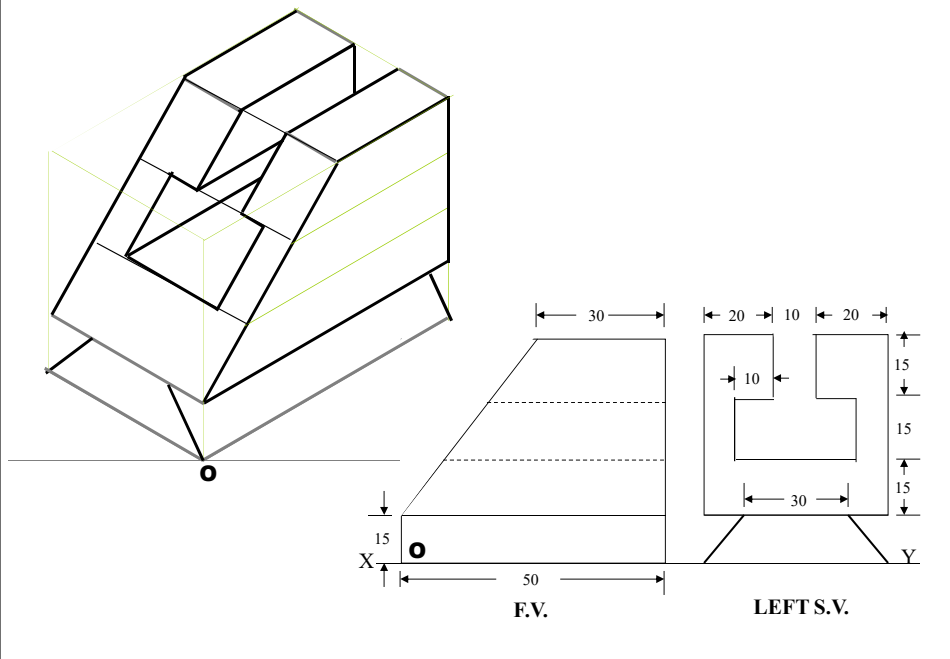
F.V. and S.V. of an object are given. Draw its isometric view.



NOTE THE SMALL CHZNGE IN 2ND FV & SV.
DRAW ISOMETRIC ACCORDINGLY.



F.V. and S.V. of an object are given. Draw it's isometric view.



F.V. and S.V. of an object are given. Draw it's isometric view.

